

Brain Tumor MRI Classification: Comparing a Frozen Autoencoder, ResNet-50, and ViT-B/16 Without Synthetic Data

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Abstract—Automated classification of brain tumors from MRI scans can support faster and more consistent diagnosis, but deep learning models for this task typically require large, balanced, annotated datasets that are hard to obtain in the medical domain. A recent reference work proposed a hybrid pipeline combining an Autoencoder, a class-conditional Generative Adversarial Network for synthetic MRI generation, and a fine-tuned Swin Transformer classifier, reaching up to 99.54% test accuracy. In this project, we reproduce the classification stage of that pipeline without the synthetic-data-generation component, comparing classification backbones under one shared data pipeline on the same public four-class brain tumor MRI dataset: a convolutional Autoencoder whose frozen encoder feeds either a linear or an MLP head, a ResNet-50 fine-tuned from ImageNet weights (with a from-scratch variant as an ablation), and a fully fine-tuned ViT-B/16. The fully fine-tuned backbones substantially outperform either frozen-encoder classifier; switching the CAE’s head from linear to a small MLP recovers +10.88 points of test accuracy from the same frozen representation, while removing ImageNet pretraining costs ResNet-50 only -1.56 points, indicating it mainly aids convergence speed rather than final accuracy. We used per-epoch training/validation loss curves to identify each fine-tuned model’s early-stopping point, and Grad-CAM to verify that the pretrained ResNet-50 attended to the tumor region itself, including a per-error reading of its misclassifications. Our results clarify how much of the reference paper’s performance stems from full backbone fine-tuning, as opposed to its synthetic-data-augmentation stage.

Index Terms—Convolutional Neural Networks, Vision Transformers, Autoencoders, Transfer Learning, Medical Image Classification, Early Stopping.

I. INTRODUCTION

Brain tumors are among the most life-threatening neurological conditions, and their prognosis depends heavily on early and accurate diagnosis. Magnetic Resonance Imaging (MRI) is the primary non-invasive tool radiologists use to detect and classify brain tumors, producing high-definition images without harming the patient. Manual reading of MRI scans is nonetheless time-consuming, requires highly specialized expertise, and is subject to inter-observer variability, making automated tumor classification from MRI an active research area, promising faster and more consistent diagnostic support.

The main problem for deep learning in this field is the scarcity of large, balanced, high-quality annotated datasets, since classic augmentation strategies (flips, rotations, crops) only produce limited additional diversity. Recent work has therefore explored combining generative models with modern classification backbones to enlarge the effective training

set and improve learned representations. In particular, Al-muhaimeed et al. [1] recently proposed the GAS model, a hybrid pipeline that couples an Autoencoder (AE), a class-conditional Generative Adversarial Network (cGAN) for synthetic MRI generation, and a fine-tuned Swin Transformer classifier (Sec. II details the pipeline and its accuracies on two public brain-tumor MRI datasets).

This project takes [1] as its reference paper and starting point. Rather than reproducing the full generative pipeline (AE + cGAN + Swin Transformer), which is computationally heavy and whose main accuracy gains stem from the synthetic-data augmentation stage, we focus on a narrower and more tractable question: how much of the reference paper’s classification performance can be recovered without any synthetic data generation, using only the original real images? What is the relative performance of different backbone choices (a small autoencoder-based feature extractor, a fine-tuned convolutional network, and a fine-tuned vision transformer) when subjected to an identical data and training pipeline? We use the same public dataset and the same class set (glioma, meningioma, pituitary tumor, no tumor) as the reference paper [1].

The contributions of this report are:

- One targeted comparison between a linear vs. MLP classification head on top of the same frozen CAE encoder, in order to quantify how much accuracy comes from head capacity;
- One targeted ablation for an ImageNet-pretrained vs. from-scratch ResNet-50, quantifying how much accuracy comes from pretraining;
- One direct comparison between two fully fine-tuned backbones, ResNet-50 and ViT-B/16, isolating the effect of architecture family (convolutional vs. self-attention);
- A Grad-CAM [2] interpretability analysis of the pretrained ResNet-50 classifier, showing which image regions drive its predictions, including a per-error reading of its two misclassified cases;
- An explicit train/validation loss-curve analysis for all four primary models, used to identify, for each architecture, the epoch at which early stopping would have preserved the best-generalizing weights.

This report is structured as follows. Sec. II reviews the reference GAS model; Sec. III and Sec. IV describe the

processing pipeline and dataset; Sec. V details the three learning architectures (4 different models); Sec. VI presents results, ablation and comparisons; Sec. VII concludes.

II. RELATED WORK

Our reference paper is Almuhaimeed et al. [1], which proposes the GAS model for brain tumor MRI classification: a convolutional Autoencoder (AE) whose encoder helps the Discriminator of a class-conditional GAN (cGAN) distinguish real from synthetic images (rather than feeding the final classifier directly), a cGAN Generator that produces 500 synthetic MRI images per class to counter class imbalance, and a fully fine-tuned Swin Transformer (Swin-Tiny), trained on the combined real-plus-synthetic set as the final classifier. On the Kaggle Brain MRI dataset [3] (four classes, 80/10/10 split), an AE + Swin configuration without cGAN augmentation already reaches 98.93% test accuracy, and adding synthetic images pushes this to 99.54%, a comparatively small gain with most of the performance already coming from the fine-tuned Swin Transformer alone. Notably, the Autoencoder never actually supplies features to the final classifier in that pipeline: it is trained purely for reconstruction, and reused only inside the GAN’s Discriminator, making it an auxiliary, GAN-facing component rather than itself a classification backbone.

Our project asks a complementary question by making the Autoencoder’s encoder itself the classification backbone: we train a CAE on the same dataset for reconstruction, freeze its encoder, and attach a classification head (a linear probe or a small MLP for comparison) on top of the resulting latent representation, exactly the “unsupervised-features-as-classifier” design that is often assumed but not actually used in [1]. We compare this frozen-encoder classifier against two fully fine-tuned, ImageNet-pretrained backbones: ResNet-50 [4], a classical convolutional network also trained from scratch as a pretraining ablation, and ViT-B/16 [5], a Vision Transformer that, like the Swin Transformer used in [1], relies on self-attention but with a simpler, non-hierarchical windowing scheme.

III. PROCESSING PIPELINE

Fig. 1 shows the overall processing pipeline. A raw MRI image is resized and, for the training split only, augmented, then routed unchanged to independent classification branches detailed in Sec. V. Each branch is trained and evaluated independently on the same splits for a fair comparison; Grad-CAM [2] is additionally applied to the pretrained ResNet-50 branch (Sec. VI-C).

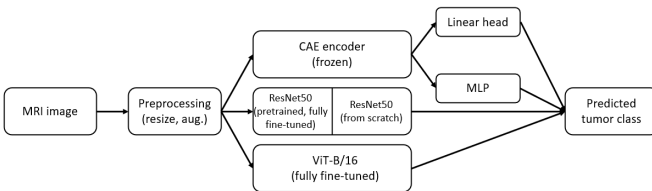


Fig. 1: Overview of the processing pipeline.

IV. DATASET AND PREPROCESSING

We use the Kaggle Brain Tumor MRI dataset [3], the same public dataset used in our reference paper [1]. It contains 7200 MRI scans labeled into four classes: glioma, meningioma, pituitary tumor, and no tumor. They are collected from axial, sagittal, and coronal slices, and originally distributed as separate Training/ and Testing/ folders. Fig. 2 shows one representative sample per class.

Unlike [1], which merges the original training and testing folders and re-splits everything from scratch, we keep the official Testing/ folder untouched and reserved exclusively for final evaluation, avoiding any test-set leakage. The Training/ folder is split into an 80% training / 10% validation subset (4,480 / 560 images), matching the same 80/10/10 ratio to the training pool as in [1] while the official Testing/ folder yields 1,600 test images. All images are resized to 224×224 . Training images are augmented with random horizontal flips ($p = 0.5$), rotations (up to 15°) and color jitter, while validation/test images only undergo resizing and normalization (ImageNet mean/std).

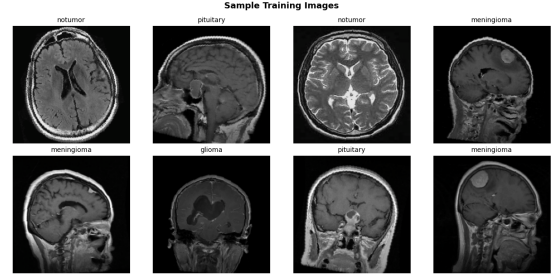


Fig. 2: Sample MRI images from the four classes after preprocessing.

V. LEARNING FRAMEWORK

A. Convolutional Autoencoder (CAE)

The CAE consists of a convolutional Encoder and a symmetric transposed-convolutional Decoder [6]. The Encoder applies four convolution-batchnorm-ReLU-maxpool blocks (channel widths $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$), halving the spatial resolution at each stage from 224×224 down to 14×14 , followed by a linear projection to a 512-dimensional latent vector; this single projection layer accounts for the large majority of the Encoder’s $\sim 26.1\text{M}$ parameters. The Decoder mirrors this structure with transposed convolutions to reconstruct a $224 \times 224 \times 3$ image, with a final sigmoid activation. Encoder and Decoder together total $\sim 52.0\text{M}$ parameters. The CAE is trained for 40 epochs as a *denoising* autoencoder: Gaussian noise ($\sigma = 0.15$) is added to the Encoder’s input, while the MSE reconstruction target remains the clean (denormalized) image, forcing the bottleneck to encode robust semantic structure rather than passing pixel-level noise straight through. Training uses Adam (learning rate 2×10^{-3} , weight decay 10^{-5}) with a ReduceLROnPlateau schedule (factor 0.3, patience 1 epoch); early stopping (patience 5) was never triggered, as the rate had already decayed to near-zero by epoch ~ 36 .

Once training is complete, the Encoder is frozen and reused as a fixed feature extractor. Fig. 3 illustrates the two-stage architecture.

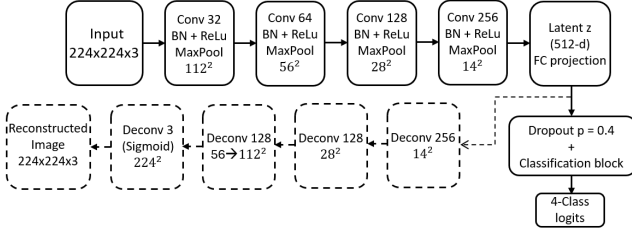


Fig. 3: CAE architecture: Stage 1 reconstruction (Encoder-Decoder) and Stage 2 classification (frozen Encoder + linear or MLP head).

- *Classification heads: linear probe vs. MLP*

Two classification heads are trained on top of the same frozen Encoder, isolating how much of the CAE branch’s accuracy is limited by head capacity rather than by the frozen representation itself. The *linear head* (dropout $p = 0.4$, Linear(512, 4), 2,052 trainable parameters) is our baseline. The *MLP head* replaces the single layer with Linear(512, 256) $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Linear(256, 4), for 132,356 trainable parameters, roughly $64\times$ more, but still under 0.3% of the frozen Encoder. Both heads are trained identically: 25 epochs, Cross-Entropy, Adam (10^{-3}) with cosine annealing, patience 5.

B. ResNet-50

The second branch uses a ResNet-50 [4] backbone (~ 23.5 M parameters, replacing the original 1000-way head with a dropout layer $p = 0.4$ followed by a linear layer producing 4 class logits): a 7×7 stride-2 convolutional stem and max-pooling reduce the input to 56^2 , followed by four stages of bottleneck residual blocks (3, 4, 6, and 3 blocks, respectively) that progressively halve the spatial resolution while doubling the channel width, from $56^2 \times 256$ down to $7^2 \times 2048$. A global average pooling layer then reduces the final feature map to a 2048-dimensional vector. Fig. 4 illustrates the backbone and the replaced head. Two variants of this backbone are trained, isolating how much of its performance comes from ImageNet pretraining versus the architecture itself.

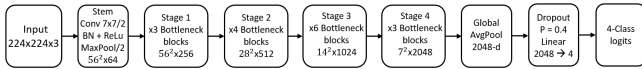


Fig. 4: ResNet-50 architecture, shared by Case 1 (pretrained) and Case 2 (scratch).

1) *Case 1 – ImageNet-pretrained*: This is our primary Stage 2 model, the one we use for the main comparison against the CAE and ViT branches. We only arrived at this specific setup because an earlier attempt at fine-tuning overfit rapidly, which pushed us to make two changes. First, we froze the

stem and the first residual stage, since their filters capture only low-level, generic patterns that rarely need to change for a medical-imaging task. Second, instead of using a single learning rate for everything else, we gave each remaining layer its own: 10^{-5} for Stage 2, 3.3×10^{-5} for Stage 3, and 10^{-4} for Stage 4 and the head. This way, the deeper, more task-specific layers can adapt faster, while the earlier ones mostly just stay close to their pretrained values. We train this configuration with the AdamW optimizer (weight decay 10^{-5}), a cosine-annealing schedule, for up to 20 epochs, with a patience of 5.

2) *Case 2 – trained from scratch*: As a control, the identical architecture is also trained with randomly initialized weights, no layer freezing, and a single higher learning rate (10^{-3} , plain Adam) applied to all layers over a longer 30-epoch budget to compensate for slower convergence, and early stopping with patience 5 to avoid overfitting. This case is not meant to outperform Case 1; it exists solely to quantify the contribution of ImageNet pretraining given our comparatively small training set (4,480 images). It is evaluated only on final test accuracy (Sec. VI-B), without the full confusion-matrix/Grad-CAM treatment given to our primary models.

C. ViT-B/16

The third branch uses a ViT-B/16 [5] backbone (~ 85.8 M parameters) pretrained on ImageNet. The image is split into 16×16 patches, linearly embedded into 768-dimensional tokens, prepended with a learnable [CLS] token, and processed through 12 Transformer encoder blocks (multi-head self-attention with 12 heads, followed by an MLP), each wrapped in residual connections. The final [CLS] token representation is passed through a dropout layer ($p = 0.3$) and a linear classification head. As with ResNet-50 Case 1, the entire network is fine-tuned end-to-end, using the AdamW optimizer with a small learning rate (2×10^{-5} , since large pretrained Transformers are more sensitive to aggressive updates), weight decay 0.01, a two-epoch linear warm-up, and a cosine decay for the remaining epochs. Fig. 5 illustrates the patch embedding, the repeated Transformer encoder block, and the classification head.

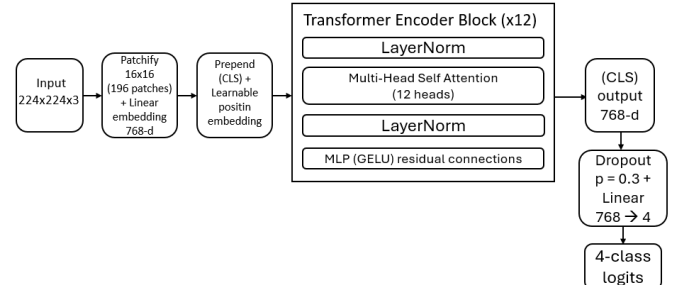


Fig. 5: ViT-B/16 architecture: patch embedding, 12 Transformer encoder blocks, and classification head.

D. Common training protocol

All supervised classifiers (the CAE’s linear and MLP heads, ResNet-50 Case 1 and Case 2, and ViT-B/16) are trained

with Cross-Entropy loss, recording per-epoch training/validation loss and accuracy. Early stopping stops training once validation loss fails to improve for a fixed number of consecutive epochs: 5 for every branch except ViT-B/16, which uses 3 (rationale in Sec. V-E). We reload each model’s best checkpoint (lowest validation loss), rather than its final-epoch weights, for all test-set evaluations, Grad-CAM, and figures in Sec. VI. Checkpoints and histories are cached to persistent storage for reproducibility.

E. Hyperparameter Selection and Tuning Rationale

Hyperparameters were chosen from standard literature defaults for each backbone family and adjusted empirically by monitoring per-epoch training/validation loss and accuracy curves. The rationale for each branch’s learning rate, scheduler, and, for ResNet-50, its layer-freezing strategy is discussed alongside its architecture in Sec. V. Dropout is set to 0.4 for both CAE heads and for ResNet-50, but only 0.3 for ViT-B/16. This isn’t about how many parameters each model has, it’s about how much each model’s weights actually move during training. The CAE heads start from scratch. ResNet-50’s later layers are fine-tuned at a fairly high learning rate (10^{-4}). Both therefore change a lot during training, so they need the stronger regularization that a higher dropout provides. ViT-B/16 is different: it is fine-tuned at a much lower learning rate (2×10^{-5}), starting from strong ImageNet pretraining. Its weights barely move, and that pretraining already acts as a form of regularization on its own. That is why a lower dropout is enough for ViT-B/16. Batch size (32) and resolution (224×224) are shared across branches for a fair comparison and to fit the largest model (ViT-B/16) in GPU memory. Data augmentation (horizontal flips, rotations up to 15° , mild color jitter) is kept conservative rather than aggressive, since MRI orientation is diagnostically meaningful and stronger distortions could produce anatomically implausible images.

The early-stopping patience was derived empirically rather than fixed a priori: Fig. 7 shows validation loss rising gradually for most branches, motivating a patience of 5 to avoid stopping on transient fluctuations without wasting compute, whereas ViT-B/16’s validation loss turns upward sooner and more sharply, motivating a shorter patience of 3.

VI. RESULTS

We first check whether the CAE has learned a meaningful representation before using its encoder as a fixed feature extractor.

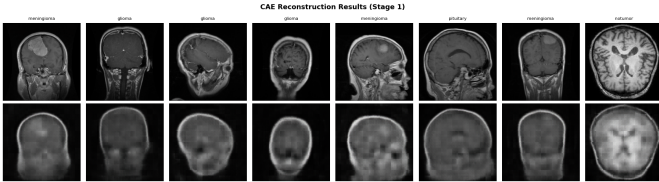


Fig. 6: Original (top row) vs. reconstructed (bottom row) MRI images produced by the trained CAE.

Fig. 6 shows original test images next to their reconstructions. The reconstructed images preserve overall brain shape

and tumor location, though fine texture detail is smoothed out, which is expected given the bottleneck imposed by the 512-dimensional latent space.

A. Train/validation loss curves and early stopping

Before deciding on an early-stopping criterion, we first ran each of the four supervised classifiers (CAE linear head, CAE MLP head, ResNet-50 Case 1, ViT-B/16) for their full, fixed epoch budget, recording the training and validation loss at every epoch.

This exploratory analysis is what motivated the early-stopping patience adopted in the final pipeline (Sec. V-E). Since patience is measured from the best epoch itself, enabling early stopping only changes how many (wasted) epochs are trained after the optimum is reached.

Fig. 7 shows all four curves side by side, with the epoch of lowest validation loss marked with a star and the region past it shaded, i.e., the region in which the model is, on net, overfitting. The restored best-checkpoint weights, and therefore the final test-set results reported below, are unaffected.

For both CAE classification heads, the validation loss is, for most of the training, lower than the training loss: it’s a direct consequence of the asymmetry between the two evaluation conditions (augmentation and dropout are active only during training), not evidence of better generalization. This is consistent with the frozen encoder and the heads’ limited capacity: with so few trainable parameters, there is little room for either head to overfit in the first place. The lowest validation loss is reached at epoch 24 of 25 for the linear head, and at epoch 23 of 25 for the MLP head, both very late in their budget, consistent with a head still finding marginal gains in an already-fixed representation rather than quickly overfitting to it.

ResNet-50 (Case 1) and ViT-B/16, in contrast, are fully fine-tuned with far more trainable parameters, and both show the more familiar pattern of validation loss eventually turning upward while training loss keeps decreasing: ResNet-50’s best epoch is 12/20, and ViT-B/16’s is 3/15, after which it early-stops at epoch 6 (patience 3). ViT-B/16 converges so quickly (96.4% validation accuracy after just the first epoch) that most of its 15-epoch budget would otherwise be spent overfitting.

B. Test-set performance

Fig. 9 compares the final test-set accuracy of our four models, each evaluated using the checkpoint corresponding to its best validation epoch (Sec. VI-A).

ResNet-50 and ViT-B/16 substantially outperform the frozen-encoder CAE classifier, regardless of which CAE head is used. This gap suggests that a frozen, unsupervised Autoencoder representation is far less competitive as a classification backbone than a fully fine-tuned, task-adapted network of either family (convolutional or Transformer-based), despite the CAE encoder being by far the cheapest of the three architectures to train.

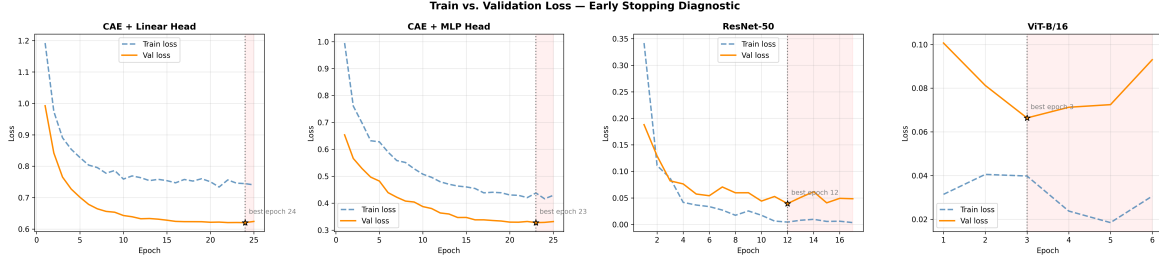


Fig. 7: Training vs. validation loss for the four supervised classifiers.

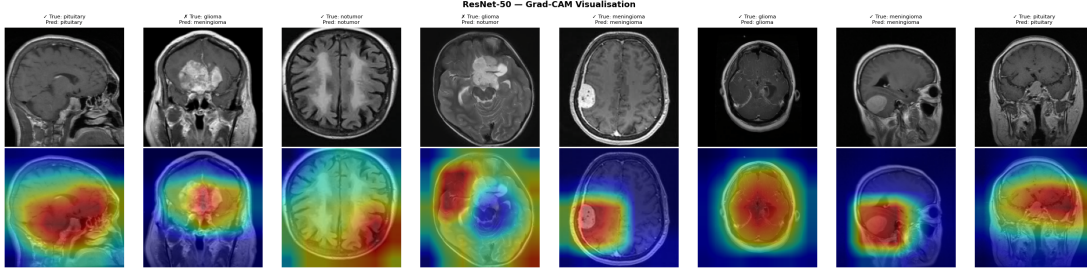


Fig. 8: Grad-CAM overlays for the ResNet-50 (Case 1) classifier on eight test images.

TABLE 1: Test-set accuracy per model.

Model	Test accuracy (%)
CAE (frozen encoder + linear head)	72.88
CAE (frozen encoder + MLP head)	83.75
ResNet-50 (Case 1, pretrained)	95.25
ViT-B/16 (fully fine-tuned)	95.06

Between the two fully fine-tuned backbones, ResNet-50 and ViT-B/16 land within 0.19 percentage points of each other (95.25% vs. 95.06%, Tab. 1), about 3 images out of 1,600, well within normal sampling noise for a test set this size, and the same ordering already visible at validation (98.93% vs. 98.21%). The two architectures are thus effectively tied, suggesting that, for this dataset size and task, full fine-tuning of a pretrained backbone matters considerably more than the specific choice between convolutional and self-attention-based feature extraction. This result is consistent with the reference paper’s [1] own choice of a fully fine-tuned Swin Transformer as its main classifier.

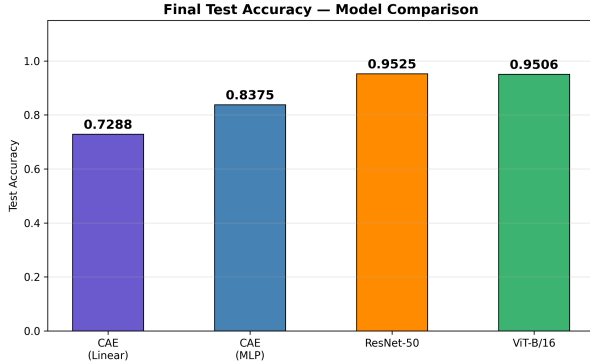


Fig. 9: Test-set accuracy of the four classifiers.

1) *CAE heads comparison:* Since both CAE heads share the same frozen Encoder and data, their gap isolates how

much accuracy is limited by the head rather than the representation. Replacing the linear head with a one-hidden-layer MLP (132,356 vs. 2,052 parameters) raises test accuracy from 72.88% to 83.75%, a +10.88 points gain from the head alone, showing the frozen Encoder already captures more class-relevant information than a linear boundary can exploit.

2) *ImageNet pretraining ablation:* ResNet-50 Case 2 (from scratch, Sec. V-B2) reaches 93.69% test accuracy, 1.56 points below Case 1’s 95.25%. Pretraining also converges faster and more stably: Case 1 climbs near-monotonically to its best epoch (12/20), while Case 2 swings hard early on and only reaches its best epoch at 18/30, early-stopping at 23. These results suggest that here pretraining mainly buys convergence speed rather than a large accuracy advantage. This confirms our choice to adopt Case 1.

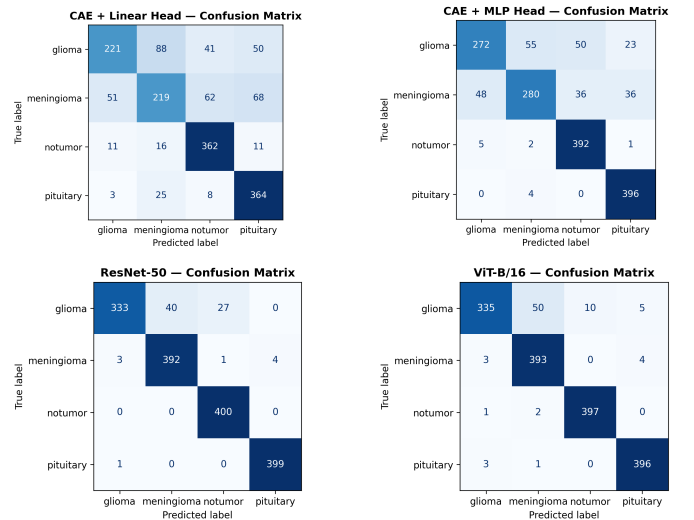


Fig. 10: Test-set confusion matrices for the four primary classifiers.

Fig. 10 shows the per-class confusion matrices for all four models on the test set. Glioma is the hardest class to classify correctly, but recall on it improves steadily as model capacity increases: 0.55 with the CAE linear head, 0.68 with the MLP head, and 0.83–0.84 for the two fully fine-tuned backbones. However, it is still the weakest class, pointing to a genuine data/task difficulty (gliomas are visually the most heterogeneous tumor type) that more capable models only partially overcome. Low glioma recall consistently pairs with high glioma precision and lower meningioma precision, indicating misclassified gliomas are predominantly predicted as meningioma rather than scattering randomly. This weakness pairs with the results in the reference paper [1], attributed there to visual similarity between the two tumor types. The CAE branch makes this especially clear: switching from linear to MLP head lifts recall specifically on this pair (0.55→0.68 glioma, 0.55→0.70 meningioma) while barely moving notumor/pituitary, already ≥ 0.90 with either head.

C. Interpretability: Grad-CAM on ResNet-50

Grad-CAM analysis addresses a question test accuracy alone cannot answer: whether the gap between the fully fine-tuned backbones and the frozen-encoder CAE (Sec. VI-B) reflects genuinely more meaningful features, or merely a better-calibrated boundary on the same superficial cues. We apply it only to ResNet-50 Case 1, for the reasons already discussed (Sec. V-B2). ViT-B/16 would instead require an attention-based method such as attention rollout (future work, Sec. VII), and the CAE’s much lower accuracy would make such an analysis far less informative.

We apply Grad-CAM to the last residual block and overlay the resulting heatmaps on eight test images spanning all four classes (Fig. 8), with true/predicted labels. In the six correctly classified cases, heatmaps consistently overlap with the tumor itself, indicating the network localizes the pathology rather than exploiting background cues. The two misclassified cases, both true gliomas, are more informative: the image predicted as meningioma still shows a heatmap over the same lesion, suggesting the network detected the abnormality but assigned the wrong class. Instead, the image predicted as no-tumor shows a diffuse heatmap that fails to concentrate on the lesion, a genuine localization failure and a clinically more consequential error, since it is a missed detection rather than a subtype confusion.

This per-error analysis is absent from the reference paper [1]: beyond confirming that fine-tuning attends to the tumor region in most cases, it shows that ResNet-50’s errors are not all equivalent, a distinction that the aggregate accuracy in Tab. 1 cannot make alone.

VII. CONCLUDING REMARKS

We compared four brain-tumor MRI classifiers: a frozen-encoder CAE with a linear and one with an MLP head, an ImageNet-pretrained ResNet-50 (fine-tuned and trained from scratch), and a fully fine-tuned ViT-B/16, following the same dataset, classes, and split philosophy as the reference GAS

model, but omitting its cGAN augmentation stage. The two fully fine-tuned backbones tie at about 95% test accuracy, far above either CAE head. The MLP head alone recovers +10.88 points over a linear one, while removing ImageNet pretraining costs ResNet-50 only −1.56 points, showing that pretraining here mainly buys convergence speed rather than a higher accuracy ceiling. Since ResNet-50 matches ViT-B/16’s accuracy with roughly a quarter of its parameters ($\sim 23.5\text{M}$ vs. $\sim 85.8\text{M}$), it appears to be the better accuracy-to-size trade-off of the two backbones. This is a potentially meaningful advantage for medical deployment, where smaller models generally mean lower memory and compute requirements, making clinical deployment more practical. Grad-CAM further confirmed that the pretrained ResNet-50 attends to the tumor region in most cases, including most of its errors.

These results matter wherever labeled medical data is scarce: a frozen-encoder classifier is a poor default, and full fine-tuning of a pretrained backbone matters more than the specific choice of architecture. The main limitation of this work is the missing cGAN stage, which prevents a direct comparison with the reference paper’s headline accuracy.

A key lesson was how much small implementation choices can dominate a fine-tuning outcome: our initial, naively fine-tuned ResNet-50 overfit sharply, which we resolved through layer freezing and discriminative learning rates.

Among all architectures, the CAE proved to be the most challenging to implement from scratch due to the need to balance its reconstruction objective and stabilize latent representations, whereas the other backbones benefited from established pretrained weights.

Future work could extend attention-rollout-based interpretability to ViT-B/16, and replace our trial-and-error hyperparameter adjustment with an automated search method (e.g. random search or early-stopping-based approaches), which would explore the hyperparameter space more systematically. In addition, implementing synthetic data augmentation with a cGAN would enable a direct comparison with the headline accuracy of the reference paper.

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